

## MLA Question bank –

1. Illustrate Machine Learning and Describe the life cycle of Machine Learning.
2. Apply the concept of different types of machine learning in real world scenarios by giving examples
3. Compare and contrast supervised, unsupervised, and semi-supervised learning approaches, emphasizing their advantages and disadvantages
4. Compare AI, ML, DL.
5. Illustrate the diagrammatic representation of supervised, unsupervised, and reinforcement learning.
6. Differentiate between Training and Testing.

## Unit 2 –

1. differentiate between binary classification and multiclass classification.
2. Define Confusion Matrix and compare and contrast accuracy, precision, recall, and F1-score in the context of evaluating classifiers. Provide the following numeric values using a hypothetical dataset:

True Positive (TP) = 189

False Positive (FP) = 99

True Negative (TN) = 901

False Negative (FN) = 38

Please calculate the accuracy, precision, recall, and F1-score for the classifier using these values.

3. Compare and contrast accuracy, precision, recall, and F1-score in the context of evaluating classifiers.
4. Analyse the role of kernel methods in SVM for handling non-linearity. illustrate commonly used kernel function.
5. Apply the concept of SVM in Health care industries.
6. Apply the concept of SVM for linear regression and classification problems.
7. Discuss Perceptron with its applications

## Unit 3

1. Illustrate Regression and it's Types
2. Describe any 5 types of Regression with examples.
3. differentiate between Underfitting, Overfitting, Variance and Bias with diagrammatic representations

4. Discuss about VC Dimensions linear model. Show that the VC dimension of line is 3.
5. Illustrate Regularized Regression Techniques - Ridge Regression and Lasso Regression
6. Discuss and Apply the concept Hypothesis testing in real world applications.

#### Unit 4

1. Describe Nearest Neighbour Classification with diagram and examples.
2. Illustrate k-mean Clustering Algorithm
3. Discuss About Hierarchical clustering.
4. Describe Decision Tree Algorithms
5. Differentiate between K-mean and k-medoids clustering.
6. Discuss the concept of Entropy and Information Gain describe it's use cases.

#### Unit 5

1. Explain the concept of naïve user-based recommendation systems. How does this approach use user similarity to make personalized recommendations? Discuss any limitations of this method
2. Explain the concept of alternating least squares (ALS) in collaborative filtering. How does ALS optimize matrix factorization to predict user-item preferences?
3. List the various application related to content based recommendation system.
4. How does a naïve user-based recommendation system leverage user similarity to personalize recommendations? Evaluate its limitations.
5. List out different types of recommendation system with diagram.